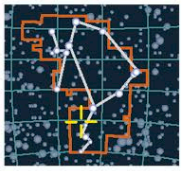
SUPERNOVA

Kepler's Star



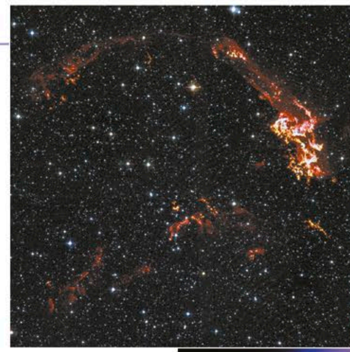
CATALOG NUMBER
SN 1604

DISTANCE FROM SUN
13,000 light-years

MAXIMUM MAGNITUDE
–2.5

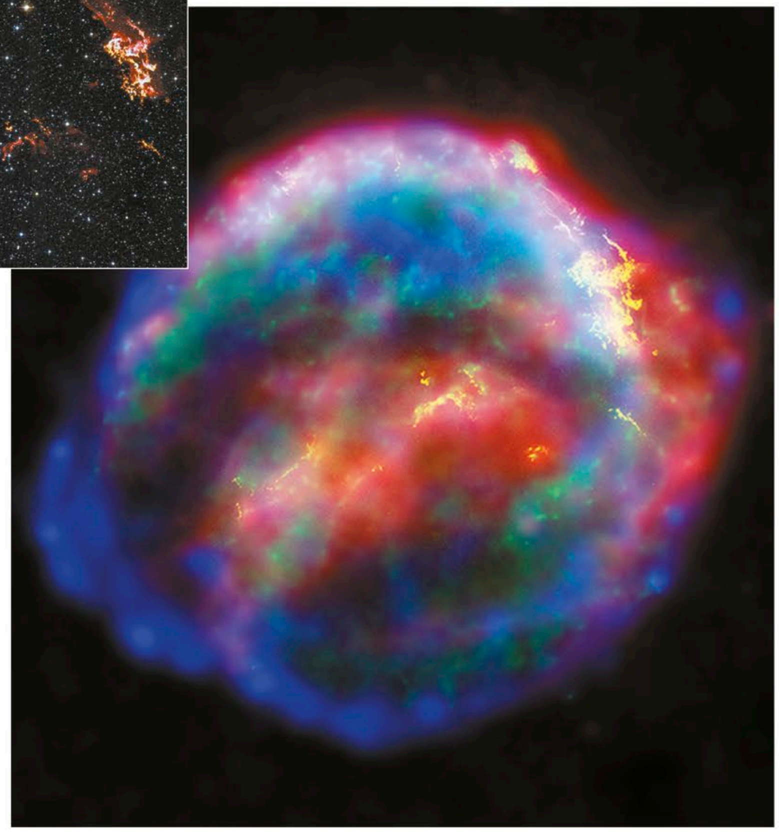
OPHIUCHUS

The last supernova explosion in the Milky Way to be observed is named after Johannes Kepler, who witnessed it in October 1604. This previously unremarkable star reached a magnitude of -2.5 and remained visible to the naked eye for more than a year. Its position is now marked by a strong radio source and, in optical light, by a wispy supernova remnant, generally known as Kepler's Star. Observations have revealed that the supernova remnant has a diameter of about 14 light-years and that the material within it is expanding at 4.5 million mph (7.2 million kph). Kepler's Star has been imaged by three of NASA's great observatories: the Hubble Space Telescope, the Spitzer Space Telescope, and the Chandra X-Ray Observatory.



VISIBLE WISPS
In this optical image, the supernova remnant appears as a faint ring of gas filaments. Having been expelled by the original explosion, this stellar material becomes heated and glows as it plows through the interstellar medium.

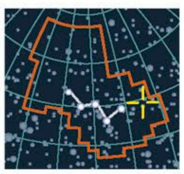
A combination of these images (right) has highlighted the remnant's distinct features. It shows an expanding bubble of iron-rich material surrounded by a shock wave, created as ejected material slams into the interstellar medium. This shock wave, shown in yellow, can also be seen optically (above). The red color is produced by microscopic dust particles, which have been heated by the shock wave. The blue and green regions represent locations of hot gas: blue indicates high-energy X-rays and the highest temperatures; green represents lower-energy X-rays.



COMBINED IMAGE
A composite picture made using images from three separate telescopes offers a view ranging from X-ray through to infrared.

SUPERNOVA

Cassiopeia A



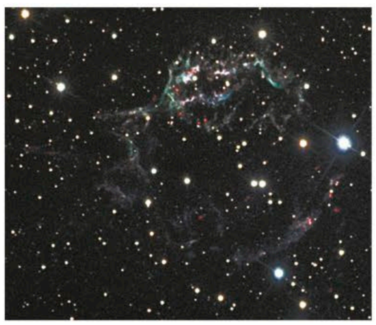
CATALOG NUMBER
SN 1680

DISTANCE FROM SUN
10,000 light-years

MAXIMUM MAGNITUDE 6

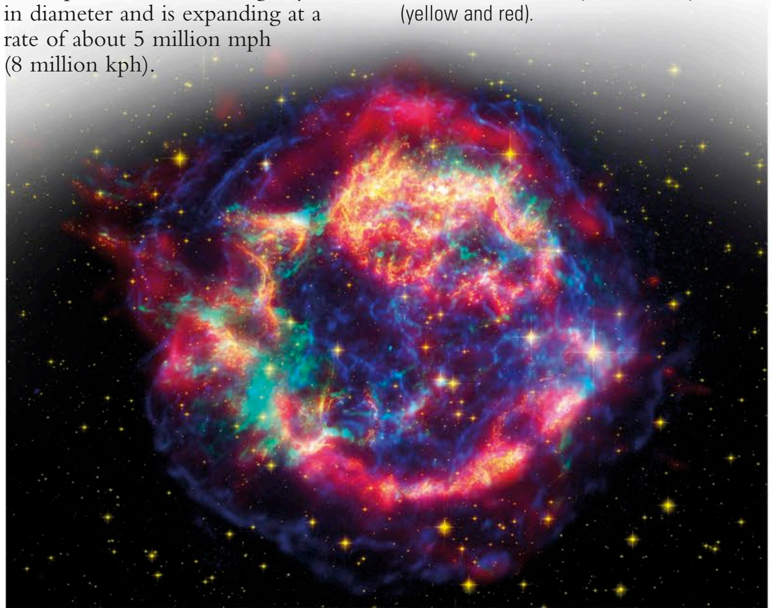
CASSIOPEIA

An intense radio source, Cassiopeia A is the remnant of a supernova explosion that occurred in the middle of the 17th century. The fact that no reports of the original explosion have been found suggests it may have been of unusually low optical luminosity. Today, Cassiopeia A is the strongest discrete low-frequency radio source in the sky (after the Sun). The radio waves are produced by electrons spiraling in a strong magnetic field. Cassiopeia A is about 10 light-years in diameter and is expanding at a rate of about 5 million mph (8 million kph).



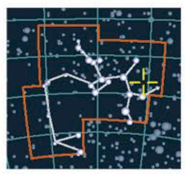
COLOR-CODED IMAGE
This Hubble Space Telescope image of Cassiopeia A's cooling filaments and knots has been color-coded to help astronomers understand the chemical processes involved in the recycling of stellar material.

SHOCK WAVES
This false-color image combines X-ray data from Chandra (blue and green) with near- and far-infrared images from the Hubble and Spitzer telescopes (yellow and red).



BLACK HOLE

MACHO 96

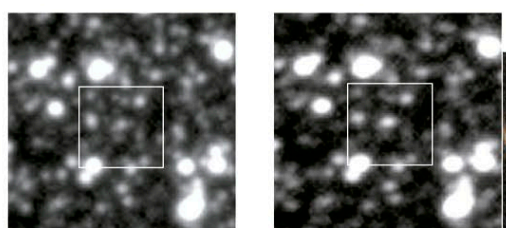


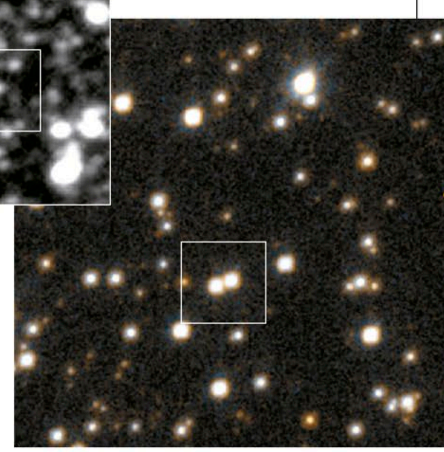
CATALOG NUMBER
MACHO 96

DISTANCE FROM SUN
Up to 100,000 light-years

SAGITTARIUS

Although we cannot see black holes, we can detect their presence by measuring their effects on objects around them. The existence of the black hole named MACHO 96 is inferred from the observed brightening of a star lying beyond the black hole caused by a process called lensing (see p.327). Through this process, the black hole's mass bends the light from the star in the same way as a lens does. The distant star is temporarily magnified, and we see a brief and subtle brightening in the star's output. The dark lensing object MACHO 96 has been calculated to be a six-solar-mass black hole that is moving independently among other stars. The chances of observing such a lensing event are estimated to be extremely slim. Therefore, astronomers monitor millions of stars every night, using computers to analyze the brightness of the stellar images captured by advanced camera systems. So far, fewer than 20 events have been detected looking toward the Large Magellanic Cloud, a nearby galaxy (see pp.310–311). MACHO 96 was initially detected by the MACHO Alert System in 1996 and subsequently monitored by the Global Microlensing Alert Network. However, it was only by studying images taken by the Hubble Space Telescope that astronomers could identify the lensed star and determine its true brightness (see below). Observations have suggested that the distant star may be a close binary system, but astronomers are still debating whether the lensing object lies in the Milky Way's Galactic Halo or in the Large Magellanic Cloud.





PASSING BLACK HOLE
Two ground-based images of a crowded star field (above) show the slight brightening of a star caused by the gravitational lensing of the passing MACHO 96. A Hubble Space Telescope image of the same area (right) resolves the star and allows its true brightness to be determined.